



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS / AGSR Division

FIRST SEMESTER 2025-2026
Course Handout (Part-II-B)

Date: 03/08/2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course

Course No. : PHA G540
Course Title : Modern Pharmaceutical Analytical Techniques
Instructor-in-Charge : Prof. S. Murugesan
Instructors : As given in time table

No of Lecture Hours per Week: 03 No. of Practical Hours per Week: Three (1 X 3)

1. Course Description:

This course is a laboratory oriented course and covers various analytical aspects like principles of sample preparation, method development for analysis and characterization of Active Pharmaceutical Ingredients (API) and formulations, using techniques such as High Performance Liquid chromatography (HPLC), bio-chromatography, size exclusion, affinity, chiral, fast protein liquid chromatography (FPLC). Characterization of nanopharmaceuticals using Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Dynamic Light Scattering (DLS) techniques. ¹H and ¹³C Nuclear Magnetic Resonance (NMR) spectroscopy in structural characterization, Mass Spectrometry and its applications, Elemental analysis, Optical, chiro-optical techniques in structure elucidation.

2. Scope of the Course:

The course is aimed at exposing the students to modern analytical techniques in relevance to pharmaceutical industries. Training with respect to the handling and operation of sophisticated instruments, interpretation of results obtained for the characterization of molecules, their qualitative and quantitative evaluation are emphasized.

Objectives of the Course:

Exposure to principles, instrumentation, calibration and operation of analytical instruments.

Obtaining spectral data, their interpretation, evaluation of qualitative and quantitative aspects of drugs and their estimation.

3. Learning Outcomes:

Handling of analytical instruments for performing qualitative and quantitative analyses of drugs.

Design and development of suitable analytical methods for drugs based on their physico-chemical characters meeting the standard requirements.

Ability to interpret the analytical graphs and spectra for drug identification, characterization, evaluation and decision making.

4. Text Books:

1. Yuri Kazakevich and Rosario Lofrutto. HPLC for pharmaceutical Scientists. John Wiley and Sons, Inc., N.Y., 2007.
2. Robert M Silverstein, Francis X Webster -"Spectroscopic Identification of Organic Compounds", Eighth Edition, John Wiley and Sons, Inc., N.Y., 2014.
3. Robert Kelsall, Ian Hamley and Mark Geogegan. Nanoscale Science and Technology. John Wiley and Sons, Inc., N.Y., 2005.

5. Reference Books:

1. Handbook of Analytical Techniques. Edited by Helmut Gunzler and Alex Williams, Vol I and II, Wiley VCH, 2001.
2. Comprehensive Chiro-optical Spectroscopy by Nina B. Prasad LP, Koji N, Robert WW. John Wiley and Sons, Inc., N.Y 2012.
3. Organic Elemental Analysis by Wolfgang J. Kristen. Academic Press. 1983.



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4. A.H. Beckett and J.B. Stenlake - "Practical Pharmaceutical Analysis", Fourth Edition, CBS Pub., New Delhi., Vol. I & II, 1988.
5. Introduction to Spectroscopy by Donald L. Pavia, Gary M. Lampman, George S. Kriz, James A. Vyvyan, Cengage Learning, 2008.
6. Williard H.H., et al., "Instrumental Methods of Analysis", CBS Pub., New Delhi, 7th ed., 1988.
7. Parimoo P, - "Instrumental Methods of Analysis", CBS Pub., New Delhi, 1988.
8. Ewing, Galen Wood. - "Instrumental Methods of Chemical Analysis", 5th Edition, Mc.Graw-Hill, Inc., N.Y., 1985.
9. Skoog. Holler. Nieman. "Principles of Instrumental Analysis", 5th Edition, Thomson Publishers, NewDelhi, 2006.

6. Course Plan:

a. Lecture Plan

Lect. No.	Learning Objectives	Topics	Chapter in the Text Book
01	Introduction of different analytical techniques to understand the scope and applications in the area of Pharmacy – broad discussion	Introduction to Modern Pharmaceutical Analytical techniques.	Introductory chapters of T and R books
02-04	Understand the principles of sample preparation, method development for analysis and characterization of Active Pharmaceutical Ingredients (API) and formulations.	General principles of sample preparation, method development for analysis and characterization of Active Pharmaceutical Ingredients (API) and formulations using various chromatographic techniques.	T1-6 and R1 -6
05-12	Learn the theory, instrumentation, data analysis followed by applications in analytical method development of API and formulations	High Performance Liquid chromatography (HPLC), HPTLC, GC, SCFC, bio-chromatography, size exclusion, affinity, chiral, fast protein chromatography (FPLC), Basics of Hyphenated techniques.	T1-2 to 7; T1-22 and R1-12
13-20	Characterization of nanopharmaceuticals using various modern techniques.	Techniques: Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Dynamic Light Scattering (DLS) techniques.	T3 - 2
21-28	Knowledge on ¹ H and ¹³ C NMR Spectroscopy – Theory and spectra interpretation	¹ H and ¹³ C NMR Spectroscopy, structure elucidation of organic molecules, applications.	T2-4 and 5
29-34	Learn the theory, various instrumental aspects and spectra analysis in structure characterization.	Mass Spectroscopy Basic principle, Instrumentation, Fragmentation of organic molecules, applications.	T2-2
35-40	Knowledge on elemental analysis, optical, chiro-optical techniques in structure elucidation	Elemental analysis, Optical, chiro-optical techniques.	R3 and R2

b. Laboratory Plan

Laboratory sessions will be conducted so that the registered students get hands on experience on the following analytical instruments. Certain instruments will only be demonstrated to the students depending on time availability either as a regular experiment or as a part of an assignment

1. High Performance Liquid chromatography (HPLC)
2. High Performance Thin Layer Chromatography (HPTLC)
3. Gas Chromatography (GC)



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4. Liquid Chromatography-Mass Spectrometry (LC-MS)
5. Gas Chromatography - Mass Spectrometry (GC-MS)
6. Zeta Sizer Nano Series (Dynamic Light Scattering Tech)
7. Differential Scanning Calorimeter (DSC)
8. Thermogravimetric Analyzer (DTA-TGA)
9. Field Emission Scanning Electron Microscope (FE-SEM)
10. Atomic Force Microscope (AFM)
11. ^1H and ^{13}C Nuclear Magnetic Resonance Spectrometer (NMR)
12. High Resolution Mass Spectrophotometer (HRMS)
13. Elemental Analyzer
14. Powder XRD
15. Spectrophotometer
16. FT-IR, ATR-IR Spectrophotometer

Note:

1. Modifications / adjustments would be made in the theory / experimental pattern / part, if necessary as and when situation arises.
2. Students should maintain and bring updated record note-books for every practical class.
3. Make-ups for practical are not always possible. However, depending upon the genuineness of the situations, students may be permitted to perform back-log experiments, if any instructor is free, outside regular class hours.
4. It is imperative that all students come prepared for the experiment in the next turn completing all pending work concerned with the previous experiment. Adequate preparation for the practical in terms of principles and operation of the instrument as per instructions, familiarization of protocols involved, outside class hours, is mandatory.

7. Evaluation Scheme:

EC No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Remarks
1.	Mid-semester Exam	90 min	30	05/10 AN2	CB
2.	Continuous assessment including practicum component		35		CB & OB (or) CB/OB
3.	Comprehensive Exam	180 min	35	03/12 AN	CB & OB (or) CB/OB

Note: Continuous assessment will be based on theory covered in class. Topics and number will be announced in the class. It will be based on home assignments, tutorials, quiz, seminar, laboratory, viva- voce and class participation. Laboratory component may include assignments, which will be practical or theoretical type that would include interpretation of IR, NMR, Mass spectral, Elemental data - characterization of compounds, etc., besides identification and estimation of known and unknown drugs in given samples based on experiments performed.

8. **Chamber consultation hours:** To be announced in class.
9. **Notices:** Notices concerning the course will be communicated through mail / displayed on department NB.
10. **Make-Ups:** Generally, make-up will be considered for regular students only and Make-Ups are not given as a routine. It is solely dependent upon the GENUINENESS OF THE CIRCUMSTANCES under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge. IN NO CASE THE MAKE-UP APPLICATION BE SLIPPED INSIDE THE CHAMBER OF THE INSTRUCTOR-IN-CHARGE. However, the decision of the Instructor-in-Charge in the above matter will be final.
11. **Academic Honesty and Integrity Policy:** Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable. **Attendance** is compulsory as per Institute policy, regularity in theory and practical classes will be decisive factor during make-up and grading, especially in borderline cases as well as make-up's.

Instructor-in-Charge
PHA G540



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